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1. There are _____ basic properties of the database relations.

- Four
- Three
- Six
- Two

2. How many types of dependencies are?

- One
- Two
- Three
- Four

3. In _____ join, only selected rows of a relation are made cross product with second relation.

- Normal
- Theta
- Outer
- Semi

4. Which of the following is not a key?

- Super key
- Alternate key
- Integrity constraint key
- Primary key

5. A collection of concepts that can be used to describe the structure of a database

- Database
- DBMS
- Data model
- Data

6. Any combination of attributes with super key is ____

- Primary key
- Foreign key

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- Super key
- Candidate key

7. Binary relationships are those, which are established between
- One entity type
- Two entity type
- Three entity type
- Unlimited entities

8. _____ are called participants, when enrolled in a relationship.
- Attributes
- Relations
- Entities
- Keys

9. Incase of Context-level Diagram, the system is represented by
- One process atleast
- Two processes atleast
- One process only
- Any number of processes

10. Collection of raw facts is called _____.
- Information
- Object
- Output
- Data

11. A software package designed to store and manage database
- Database
- DBMS
- Data model
- Data

12. In _____ join common attributes appear twice in the output with selected rows. a.
- Semi join
- Natural join
- Equi join

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- Normal join

13. Within a table, each primary key value _____.

- Is a minimal super key
- Is always the first field in each table
- Must be numeric
- Must be unique

14. _____ is not an operation of relational algebra.

- Addition
- Union
- Cartesian product
- Set difference

15. Candidate keys which are not selected as primary key are called _____

- Secondary keys
- Foreign keys
- Alternate keys
- Composite keys

16. External scheme evolves as user needs are _____ over the time.

- Ends
- Modified
- Switch
- Stable

17. An attribute that is the collection of some other attribute(s) is known as _____ attributes.

- Multi-valued
- Single-valued
- No-valued
- Composite

18. Which of the following is correct regarding Dataflow diagram?

- Single DFD is required to represent a system
- The dataflow must be bidirectional

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- Created at increasing levels of detail
- Used to represent the relationships among the external entities

19. _____ are those attributes that are a combination of two or more than two attributes.

- Null value
- Primary key
- Composite attributes
- Default value

20. Controlling redundancy in a database management system DOES NOT help to

- avoid duplication
- avoid unnecessary wastage of storage space
- avoid unauthorized access to data
- avoid inconsistency among data

21. Which of the following concepts is applicable with respect to 3NF?

- Full functional dependency
- Any kind of dependency
- Transitive dependency
- Partial functional dependency

22. Identify the operation which is NOT one of the parts of the five basic set operations in relational algebra?

- Join
- Union
- Cartesian Product
- Set Difference

23. Making a change to the conceptual schema of a database but not affecting the existing external schemas is an example of

- Physical data independence.
- Concurrency control.
- Logical data independence.

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- Functional dependency

24. For a third normal form we concentrate on relations with one _____ key, and we eliminate transitive dependencies

- Composite
- Super key
- Primary key
- Candidate

25. The entity relationship diagram is used to graphically represent the _____ database model.

- Condensed
- Physical
- Logical
- Conceptual

26. The _____ determines that the link between two entities is optional or mandatory.

- Minimum cardinality
- Maximum degree
- Minimum degree
- Maximum cardinality

27. Select the most appropriate statement. Which of the following data model is used to design logical database design?

- Relational
- Hierarchal
- Object oriented
- Any model can be used

28. Which is not the function of Database Management System?

- A user accessible Catalog
- Concurrency Control Services
- Failure Response Service
- Support for Data Communication

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29. The conceptual database design can be transformed into any _____.

- Data model
- Database design
- E-R model
- Entity set

30. Which is the second stage in the database development process?

- Database design
- Preliminary Study
- Physical Design
- Requirement Analysis

31. Functional dependencies under interference rule reflexivity are called _____ dependencies.

- Transitivity
- Projectivity
- Trivial
- Augmentation

32. If many instances of the "Book" entity are associated with the many instances of the "Student" entity, then the relationship between them is:

- One to one
- One to many
- Many to many
- Many to one

33. Which of the following constraints enforces entity integrity?

- PRIMARY KEY
- FOREIGN KEY
- CHECK
- NOT NULL

34. Unary operations involves

- Only one relation
- Only two relations
- More than two relations

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- Not more than five relations

35. A _____ entity has a primary key that is partially or totally derived from the parent entity in the relationship.

- Strong
- Weak
- Business
- Relationship

36. In _____ appropriate data model is chosen.

- ERD
- Conceptual Database Design
- Logical Database Design
- DFA

37. Which of the following is drawback of standardization?

- Difficult to understand
- Uniformity
- Lack of uniqueness
- Easy to develop applications

38. Recursive relationship is also called _____.

- Unary
- Ternary
- Primary
- Secondary

39. The foreign key attribute, which is present as a primary key in another relation is called as

_____ of the foreign key attribute.

- Composite relation
- Home relation
- Recursive relation
- Binary relation

40. DFD does not provide us a way to represent _____

- Entities
- Decisions

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- Date stores
- Processes

41. A _____ is used to maintain a connection between the users of the database system.

- File server
- Client server
- FTP server
- Virtual server

42. Which of the following is an advantage of using the Traditional File Organization?

- Data security
- Simplicity
- Efficiency
- Time consuming

43. Which of the following is the input to Analysis Phase?

- DFD
- Logical database design
- Physical database design
- Data model

44. Which of the following most certainly implies the need for an entire table to implement?

- A ternary relationship
- A recursive relationship
- An identify relationship
- A binary relationship

45. Core of the Database Architecture is _____

- Internal level
- External level
- Conceptual level
- ANSI SPARK

46. _____ is same as equi-join with a slight difference.

- Natural join

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- Semi join
- Outer join
- Theta join

47. Conceptual database design is implemented using a _____.

- Hierarchical data model
- Semantic data model
- Relational data model
- Network data model

48. Which constraint is generally used for efficiency purposes in the data entry process?

- Integer value
- Default value
- String value
- Null value

49. A 3 NF relation is converted to BCNF by

- Removing composite keys
- Removing multivalued dependencies
- Dependent attributes of overlapping composite keys are put in a separate relation
- Dependent non-key attributes are put in a separate table

50. Each _____ of a table contains atomic/single value.

- Cell
- Row
- Column
- Record

51. A relation is said to be in BCNF when

- It has overlapping composite keys
- It has no composite keys
- It has no multivalued dependencies
- It has no overlapping composite keys which have related attributes

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52. Which one of the following is not a part of three-level architecture?

- External level
- Conceptual level
- Internal level
- Physical level

53. Which of the following is INCORRECT about naming entities?

- Singular nouns are used
- Write in capitals
- Organization specific names can be used
- Always use abbreviations as they are convenient

54. An instance is

- A particular occurance of an entity
- A special type of relation
- An attribute of an entity
- Any particular entity

55. _____ is an example of metadata

- Ali
- Address
- Person
- Student

56. Level of data at which entities or objects exist in reality is called

- Real world data
- Schema for real world data
- Meta data
- Data about data

57. Major benefits of data sharing is:

- Data of same applications placed at same place.
- Data of same applications placed at different places
- Data of different application places at same place
- Data of different applications stored at placed place

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58. Functional dependency between two attributes or sets of attributes i.e. Q and P is represented as _____.

- Q → P
- Q P
- Q * P
- Q + P

59. Which of the following is not the component of Relational Data Model?

- Integrity Constraint
- Manipulation Language
- Relation / Table
- Domain Constraint

60. The attribute that is calculated from other attribute is called _____.

- Derived attribute
- Multi-valued attribute
- Composite attribute
- Simple attribute

61. Consider the two attributes "CNIC_NO" and "Registration_NO". Both the attributes can be used for the unique identification. Now if the user selects the "CNIC_NO" as primary key, then "Registration_NO" is a ____ key.

- Foreign
- Secondary
- Primary
- Alternative

62. Which of the following constraint is included in the definition of the table?

- Default value
- Foreign key
- Null
- Domain

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63. The database deals with large amount of data. This property is called

- Database design
- Scalability
- Concurrency
- Robustness

64. Which one of the following is NOT a characteristic of meta data?

- Data about data
- Describes a data dictionary
- Self-describing
- Includes user data

65. Theta join is denoted mathematically as _____.

- R X S
- R X
- R X Ø S
- dRX RS

66. In cross reference matrix transitions are specified on _____ and database objects are specified on _____.

- Y axis, X axis
- X axis, Y axis
- X axis, X axis
- Y axis, Y axis

67. Each component of DBMS performs _____ functions.

- Similar
- Different
- Identical
- Computational

68. Which of the following data model is not record based?

- Hierarchical Data Model
- Network Data model
- Semantic Data model
- Relational Data model

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69. The basic purpose of diamond in ER diagram is to show the _____.

- Attributes
- Relationships
- Composite attribute
- Entity set

70. Duplication of data in controlled redundancy is controlled and _____.

- Deliberate
- Necessary
- Precise
- Unintentional

71. Teleprocessing is a sub-type of _____.

- Single user database
- Multi-user database
- File server
- Client server

72. In an ERD, the focus is on the _____ and the relationships between them.

- Attributes
- Entities
- Keys
- Indexes

73. The most widely used conceptual model is the _____ model.

- Implementation
- ER
- Object oriented
- Internal

74. Student and book are two entities, what is the cardinality between the two?

- Many to one
- One to many
- Many to many

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- One to one

75. RDM is a simple there is just one structure and that is _____.

- A relation or a table
- E-R model
- Attributes
- Data model

76. Consider thy key "CNIC, Name" and choose the best option that describes the key most effectively.

- The given key is a super key and alternate key.
- The given key is not a super key but a candidate key.
- The given key is a super key and candidate key as well.
- The given key is a super key but not a candidate key.

77. Normalization is a process of restricting a relation to

- Minimize duplication of data in a database
- Maximize duplication of data to ensure reliability
- Make if of uniform size
- Allow addition of data

78. Unary relationship involves a single entity it is also called _____.

- One to one relationship
- One to many relationship
- Many to many relationship
- Recursive relationship

79. Which of the following is a part of a DFD but not included in the ERD?

- The attribute having a single instance
- The external entity having a single instance
- The relationship having a single instance
- The role having a single instance

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80. The database management system (DBMS) is used to

- Store the data
- Access the data
- Manage its users
- All of above

81. Which of the following tool is used to design conceptual database design?

- SQL
- MS Access
- Oracle
- This model is independent of any tool

82. In "one to many" cardinality one instance of a relation is mapped with _____.

- Many instances of second entity type
- One instance of second entity type
- Primary key of second relation
- Foreign key of second relation

83. Cross reference matrix is developed at the _____ stage of the database.

- Analysis
- Designing
- Implementation
- Application

84. The population of the data of the organization for which the database is created is called the

_____ of the database.

- Extension
- Intension
- Model

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- Schema

85. User rights information is stored in

- Physical database
- Catalog
- Logical database
- Buffer

86. Logical data independency provides independency to change _____ model.

- External
- Logical
- Conceptual
- Internal

87. In natural join the common attribute in the output table appears _____.

- Twice
- Three times
- Four times
- Once

88. Which levels are mostly used for detailed DFD?

- Level-0, Level-1
- Level-1, Level-2
- Level-2, Level-3
- Level-3, Level-4

89. By default, a non-key attribute in a relation can have _____ value.

- Binary
- Null
- Character
- Integer

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90. The three levels architecture is useful for _____.

- Arranging the data in an organized manner
- Hiding the details of internal systems
- Breaking the data access restriction
- Storing definitions of the structures

91. With the help of _____ technique data values with the smaller sized codes can further reduce the space needed by the data for storage in the database.

- Compression
- Expansions
- Negation
- Extension

92. If an entity is linked with itself, than it is called _____ relationship.

- Binary
- Ternary
- Nary
- Recursive

93. In cross reference matrix attributes are mention on _____ and reports are mention on _____.

- Y axis, X axis
- Y axis, Z axis
- X axis, Y axis
- X axis, Z axis

94. A table can be logically connected to another table by defining a _____.

- Hyper link
- Common attribute
- Primary key

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- Super key

95. Data Redundancy means

- Security of data
- Duplication of data
- Management of data
- Recovery of data

96. Which of the following is a type of dependency?

- DBMS
- ERD
- DFD
- Referential

97. In your opinion, why relational database is widely acceptable?

- Due to its complexity
- Due to its approach
- Due to its dependencies
- Due to its strength

98. _____ is the first comprehensive complete database design.

- Logical database design
- Relational database design
- Conceptual database design
- Hierarchical database design

99. How data security can be assured in the database?

- By hiding data from all
- Through concurrent access of users
- By asking users not to share things
- Implementing encryption algorithms

100. The attributes whose value is not stored in a database is known as _____ attribute.

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- Stored
- Single value
- Multi value
- Derived

101. Logical database design, like conceptual database design is our _____

- Data model
- Weak entity set
- Strong entity set
- Database design

102. If a single instance of "student" is associated with the many instances of the "subjects" entity, then the relationship between them is:

- One to one
- One to many
- Many to many
- Many to one

103. The conceptual database design is drawn through _____.

- Data model
- ER model
- Database design
- Entity set

104. Database is a computerized representation of any organization's flow of _____ and storage of _____.

- Data, information
- Details, data
- Data, details
- Information, data

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105. In which stage of the database development process logical design is created?

- Requirement analysis
- Database design
- Physical design
- Implementation

106. Which of the following is NOT an entity?

- Employee
- Hobby
- Student
- Playground

107. After moving the determined columns from the original table to the new table while converting to 2NF, the determinate becomes the _____ of the new table.

- Primary key
- Foreign key
- Secondary key
- Composite key

108. The select operation works horizontally on the table, on the other hand the _____ operates on a single table vertically.

- Binary operator
- Project operator
- Join operator
- Select operator

109. Free standing data dictionary is created by

- DFD
- Database
- CASE tools
- Cross reference matrix

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110. A patient can be an outdoor or indoor. Analyze this statement and tell what type of constraint will be applied on patient?

- Total completeness
- Partial completeness
- Total disjoint
- Partial disjoint

111. If A → B and A → C, then A → BC. The inference Rule applies is:

- Union
- Decomposition
- Augmentation
- Reflexivity

112. Which of the following is the correct abbreviation of ERD?

- Entity relationship diagram
- Entity relation diagram
- Entity reality diagram
- Entity real distance

113. Database relation is also represented in a two dimensional structure called _____

- Row
- Table
- Entity
- Column

114. Data about data is metadata

- True
- False

115. Which of the following describes the job of a database administrator?

- Development, implementation operation of the physical database

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- Creation of the system catalog
- Monitoring and controlling database security and authorization: setting up controls to ensure the quality and integrity of data
- All of the above

116. Each table must have a _____ key.

- Primary
- Secondary
- Logical
- Foreign

117. A relational database system is based on the concept(s) of:

- A network of data records
- Nested tables
- Tables, row and columns
- A tree-like structure of data

118. A → BC and A → B, then A → C follow _____ inference rule.

- Augmentation
- Transitivity
- Projectivity
- Additivity

119. Suppose, we have two entities "shape" and "triangle", then which of the following is a correct statement?

- Shape is a super type entity and triangle is a sub type entity
- Shape is a sub type entity and triangle is a super type entity
- Shape and triangle both are super type entities
- Shape and triangle both are sub type entities

120. The entity that is NOT dependent on any other entity for its existence is known as

_____ entity.

- Weak

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- Strong
- Derived
- Dependent

121. In cross reference matrix attributes are specified on _____

- X axis
- Y axis
- Z axis
- None of the above

122. Which of the following clarify the semantics of a relationship?

- Entity
- Roles
- Relationship set
- Participants

123. How many types of cardinality is specified about relationship?

- Two
- One
- Three
- Four

124. Which of the following is NOT a component of a DFD?

- Dataflow
- Datastore
- External entities
- Relationship between external entities

125. Which of the following provides us the facility to add properties of one entity to another entity automatically?

- Abstraction
- Encapsulation
- Inheritance
- Dependencies

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126. If there is more than one relationship between two entities and we mentioned role on the relationship link, then it is called:

- Unary relationship
- Binary relationships
- Ternary relationships
- Multiple relationships

127. Which one of the following is not an advantage of DBMS

- Data consistency
- Recovery procedure is difficult
- Faster development of new applications
- Better backup and recovery procedures

128. In many to many relationship a third table is created for the relationship, which is also called as _____

- Binary relation
- Associative entity type
- Unary relationship
- Composite attributes

129. When one entity instance of another entity for its existence, then it is called _____

- Existence dependency
- Identifier dependency
- Referential dependency
- Acceptance dependency

130. Relational data model is widely accepted due to its mathematically proven foundation and its _____

- Complexity
- Simplicity
- Hierarchical structure
- Traditional structure

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131. If an entity “motor cycle” is associated with at most one “driver”, then the relationship between these two entities is:

- One to one
- One to many
- Many to many
- Many to one

132. Database application is a program which is used for performing certain operations such as

- Insertion of data
- Extraction of data
- Updating the data
- All of above

133. Creating the relationships between new tables and their predecessors through the use of foreign keys is general requirements of _____

- First normal form
- Second normal form
- Third normal form
- Fourth normal form

134. In _____ the domains of attributes of a relation are atomic, that is they consist of single units that cannot be broken down further.

- First normal form
- Second normal form
- Third normal form
- Augmentation

135. Atomicity is a feature of _____.

- 1NF
- 2NF

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- 3NF
- BCNF

136. The _____ can see entire information structure of the database.

- BA
- DBTG
- Database user
- External viewer

137. The attribute whose value is not stored in a database is known as _____ attribute.

- Stored
- Single value
- Multi value
- Derived

138. _____ are the structure defined for placing data in the attributes while designing physical design of database.

- Objects
- Variables
- Attributes
- Data types

139. Suppose a relation has five columns and ten rows, select the correct option to identify cardinality of relation.

- Five
- Ten
- Fifteen
- Twenty

140. Which feature of database provides conversion from inconsistent state of DB to a consistent state ensuring minimum data loss?

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- User accessible catalog
- Data processing
- Recovery service
- Authorization service

141. The maximum PL/SQL size of data type "VARCHAR2" in DBMS is _____ bytes

- 42767
- 52767
- 32767
- 28767

142. _____ is state of database where a "course" cannot be inserted in the table, because this course has not been registered to any "student".

- Insertion anomaly
- Deletion anomaly
- Updation anomaly
- Redundancy

143. The maximum PL/SQL size of data type "BLOG" in DBMS is_____

- 4 gigabytes
- 8 gigabytes
- 10 gigabytes
- 12 gigabytes

144. _____ states that in a relation no attributes of a primary key (PK) can have a null value.

- Referential integrity constraint
- Null constraint
- Domain constraint
- Entity integrity constraint

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145. A database state where deletion of the information about the student record deletes the course information as well is called _____.

- Insertion anomaly
- Updation anomaly
- Redundancy
- Deletion anomaly

146. An entity type is

- Defined when the database is actually constructed
- A specific type such as an integer, text, date, logical etc.
- A coherent set of similar objects that we want to store
  - data on (e.g. STUDENTS, COURSE, CAR)
  - Defined by the database designer

147. "select" and "project" are the examples of _____

- Unary operation
- Binary operation
- Ternary operation
- Nary operation

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